

6.2. Defining Planning and Addressing Spurious Correlations

A crucial consideration in defining and measuring “planning” is the potential for spurious correlations, particularly **“first-token shortcuts.”** An example illustrates this: if a model is prompted in either French or English to provide a ‘yes/no’ answer in the same language, a probe analyzing the first token’s activations might predict the final answer (“oui/non” vs. “yes/no”) simply by detecting the language of the initial token. While this shows a correlation, it doesn’t necessarily prove long-range planning in the sense of anticipating specific future content beyond what’s implied by immediate context like language choice.

Our study addresses this by (1) defining planning as the encoding of long-term attributes independently of the immediate next token, and (2) designing prompts to actively block these shortcuts. We define planning such that the hidden representations at the first token should encode both next-token information and long-term attributes, ensuring these two information types are independent – the long-term attribute shouldn’t be directly inferable from the very next token. To implement this in our experiments, we used prompt engineering. For instance, in multiple-choice tasks, we instructed models to first provide an analysis before stating their final answer. This ensures that the initial tokens (the analysis) do not inherently reveal the target attribute (the final choice), thereby helping to isolate genuine planning signals from simpler, shortcut correlations. This methodological approach is vital for ensuring that probes detect true emergent planning rather than just correlated input features.

6.3. Potential Applications of Emergent Response Planning in LLMs.

Our findings on LLMs’ emergent response planning suggest several practical applications: (1) **Pre-generation resource allocation optimization:** Probing pre-generation representations allows proactive allocation of computational resources based on anticipated response complexity and length, enhancing inference efficiency during dynamic workloads. (2) **Early-error detection:** Early detection of behavioral attributes like low confidence could enable corrective interventions (e.g., retrieval-augmented refinement) before errors propagate. Predictive awareness of content attributes (e.g., key entities or argument trajectories) might enable real-time compliance checks with safety guidelines or domain constraints. (3) **Novel user interaction paradigms:** Predicting reasoning complexity could guide task decomposition for multi-step problems, while predicting response characteristics could improve progress indicators in interactive settings. These possibilities highlight the need for robust probing methods in deployed LLMs.

6.4. Future Research Directions

Several key research directions emerge from our findings: (1) **Causal mechanisms of planning:** Research could investigate whether and how planning representations directly influence token generation (e.g., via causal intervention experiments). Establishing causality is crucial for reliably leveraging these representations and understanding LLM decision-making. (2) **Leverage planning for generation control:** Future work might explore methods to detect and utilize pre-generation attribute predictions (e.g., key content points) for real-time steering. This could enable more efficient and precise control than post-hoc correction, potentially reducing computational waste and errors by allowing early adjustments based on predicted response properties. (3) **Planning in multimodal contexts:** Exploration of whether similar planning phenomena emerge in multimodal (e.g., image+text) LLMs could be valuable. Such research may provide insights into the universality of emergent planning and how modality impacts the development of these cognitive capabilities. (4) **Planning-aware training:** Developing objectives that explicitly reward alignment between early-plan encodings and final outputs (e.g., via consistency losses) represents another avenue. This may enhance coherence in complex tasks by grounding generation in initial intent.

7. Conclusion

In conclusion, our work reveals that LLMs have emergent response planning capabilities, with prompt representations encoding global attributes of future outputs across structure, content, and behavior attributes. These findings challenge the conventional view of LLMs as purely local predictors and provide new insights into their internal mechanisms. Though we do not focus on interpretability mechanisms to explain the causal relationship of emergent response planning, our findings open promising directions for enhancing model control and transparency, potentially enabling more effective methods for guiding and predicting model outputs before generation begins.

Impact Statement

Our findings on LLM emergent planning raise specific considerations for model deployment. While these capabilities could enhance system reliability through better resource allocation and early warning mechanisms, they also present concerns when handling sensitive data, as these probing methods reveal aspects of the model’s internal thinking or decision-making process. We encourage careful evaluation of these trade-offs when implementing probing-based monitoring systems, particularly in applications involving sensitive information.

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Author Contributions

Zhichen Dong provided early inputs on the emergent response planning; proposed and ran the experimental tasks, and participated in writing all sections of the paper.

Zhanhui Zhou first pitched the research idea of emergent response planning to **Zhichen Dong**; proposed experimental tasks; made substantial writing contributions to abstract, introduction, and Section 3.

Zhixuan Liu provided valuable feedback throughout the project; **Chao Yang** and **Chaochao Lu** supervised and managed the group.

References

- Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- AI@Meta. Llama 3 model card. 2024. URL https://github.com/meta-llama/llama3/blob/main/MODEL_CARD.md.
- Anthropic. Claude 3.5 sonnet. <https://anthropic.com>, 2024. Version: claude-3-5-sonnet-20241022.
- Bachmann, G. and Nagarajan, V. The pitfalls of next-token prediction. *arXiv preprint arXiv:2403.06963*, 2024.
- Belrose, N., Furman, Z., Smith, L., Halawi, D., Ostrovsky, I., McKinney, L., Biderman, S., and Steinhardt, J. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33: 1877–1901, 2020.
- Burns, C., Ye, H., Klein, D., and Steinhardt, J. Discovering latent knowledge in language models without supervision. *arXiv preprint arXiv:2212.03827*, 2022.
- Clark, P., Cowhey, I., Etzioni, O., Khot, T., Sabharwal, A., Schoenick, C., and Tafjord, O. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv:1803.05457v1*, 2018.
- Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., and Schulman, J. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- Cornille, N., Moens, M.-F., and Mai, F. Learning to plan for language modeling from unlabeled data. *arXiv preprint arXiv:2404.00614*, 2024.
- Damani, M., Shenfeld, I., Peng, A., Bobu, A., and Andreas, J. Learning how hard to think: Input-adaptive allocation of lm computation. *arXiv preprint arXiv:2410.04707*, 2024.
- Ding, N., Chen, Y., Xu, B., Qin, Y., Zheng, Z., Hu, S., Liu, Z., Sun, M., and Zhou, B. Enhancing chat language models by scaling high-quality instructional conversations. *arXiv preprint arXiv:2305.14233*, 2023.
- Eldan, R. and Li, Y. Tinstories: How small can language models be and still speak coherent english?, 2023. URL <https://arxiv.org/abs/2305.07759>.
- Geva, M., Bastings, J., Filippova, K., and Globerson, A. Dissecting recall of factual associations in auto-regressive language models. *arXiv preprint arXiv:2304.14767*, 2023.
- Han, T., Fang, C., Zhao, S., Ma, S., Chen, Z., and Wang, Z. Token-budget-aware llm reasoning. *arXiv preprint arXiv:2412.18547*, 2024.
- Heo, J., Heinze-Deml, C., Elachqar, O., Ren, S., Nallasamy, U., Miller, A., Chan, K. H. R., and Narain, J. Do llms “know” internally when they follow instructions? *arXiv preprint arXiv:2410.14516*, 2024.
- Hernandez, E., Li, B. Z., and Andreas, J. Inspecting and editing knowledge representations in language models. *arXiv preprint arXiv:2304.00740*, 2023.
- Ji, Z., Chen, D., Ishii, E., Cahyawijaya, S., Bang, Y., Wilie, B., and Fung, P. Llm internal states reveal hallucination risk faced with a query. *arXiv preprint arXiv:2407.03282*, 2024.
- Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., de las Casas, D., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., Lavaud, L. R., Lachaux, M.-A., Stock, P., Scao, T. L., Lavril, T., Wang, T., Lacroix, T., and Sayed, W. E. Mistral 7b, 2023. URL <https://arxiv.org/abs/2310.06825>.
- Li, K., Hopkins, A. K., Bau, D., Viégas, F., Pfister, H., and Wattenberg, M. Emergent world representations: Exploring a sequence model trained on a synthetic task. *arXiv preprint arXiv:2210.13382*, 2022.

- Li, K., Patel, O., Viégas, F., Pfister, H., and Wattenberg, M. Inference-time intervention: Eliciting truthful answers from a language model. *Advances in Neural Information Processing Systems*, 36, 2024.
- Liu, P., Yuan, W., Fu, J., Jiang, Z., Hayashi, H., and Neubig, G. Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing. *ACM Computing Surveys*, 55(9):1–35, 2023.
- MacDiarmid, M., Maxwell, T., Schiefer, N., Mu, J., Kaplan, J., Duvenaud, D., Bowman, S., Tamkin, A., Perez, E., Sharma, M., et al. Simple probes can catch sleeper agents, 2024.
- Men, T., Cao, P., Jin, Z., Chen, Y., Liu, K., and Zhao, J. Unlocking the future: Exploring look-ahead planning mechanistic interpretability in large language models. *arXiv preprint arXiv:2406.16033*, 2024.
- Mostafazadeh, N., Chambers, N., He, X., Parikh, D., Batra, D., Vanderwende, L., Kohli, P., and Allen, J. A corpus and cloze evaluation for deeper understanding of commonsense stories. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 839–849, 2016.
- nostalgebraist. interpreting gpt: the logit lens. Website, 2020. <https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens>.
- Onoe, Y., Zhang, M. J. Q., Choi, E., and Durrett, G. Creak: A dataset for commonsense reasoning over entity knowledge, 2021. URL <https://arxiv.org/abs/2109.01653>.
- Pal, A., Umapathi, L. K., and Sankarasubbu, M. Medmcqa: A large-scale multi-subject multi-choice dataset for medical domain question answering. In Flores, G., Chen, G. H., Pollard, T., Ho, J. C., and Naumann, T. (eds.), *Proceedings of the Conference on Health, Inference, and Learning*, volume 174 of *Proceedings of Machine Learning Research*, pp. 248–260. PMLR, 07–08 Apr 2022. URL <https://proceedings.mlr.press/v174/pal22a.html>.
- Pal, K., Sun, J., Yuan, A., Wallace, B. C., and Bau, D. Future lens: Anticipating subsequent tokens from a single hidden state. *arXiv preprint arXiv:2311.04897*, 2023.
- Pochinkov, N., Benoit, A., Agarwal, L., Majid, Z. A., and Ter-Minassian, L. Extracting paragraphs from llm token activations. *arXiv preprint arXiv:2409.06328*, 2024.
- Qian, C., Zhang, H., Sha, L., and Zheng, Z. Hsf: Defending against jailbreak attacks with hidden state filtering. *arXiv preprint arXiv:2409.03788*, 2024.
- Rateike, M., Cintas, C., Wamburu, J., Akumu, T., and Speakman, S. Weakly supervised detection of hallucinations in llm activations. *arXiv preprint arXiv:2312.02798*, 2023.
- Sap, M., Rashkin, H., Chen, D., LeBras, R., and Choi, Y. Socialliqa: Commonsense reasoning about social interactions, 2019. URL <https://arxiv.org/abs/1904.09728>.
- Saunders, W., Yeh, C., Wu, J., Bills, S., Ouyang, L., Ward, J., and Leike, J. Self-critiquing models for assisting human evaluators. *arXiv preprint arXiv:2206.05802*, 2022.
- Saxton, Grefenstette, Hill, and Kohli. Analysing mathematical reasoning abilities of neural models. *arXiv:1904.01557*, 2019.
- Talmor, A., Herzig, J., Lourie, N., and Berant, J. CommonsenseQA: A question answering challenge targeting commonsense knowledge. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pp. 4149–4158, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1421. URL <https://aclanthology.org/N19-1421>.
- Tang, E., Yang, B., and Song, X. Understanding llm embeddings for regression. *arXiv preprint arXiv:2411.14708*, 2024.
- Taori, R., Gulrajani, I., Zhang, T., Dubois, Y., Li, X., Guestrin, C., Liang, P., and Hashimoto, T. B. Stanford alpaca: An instruction-following llama model. https://github.com/tatsu-lab/stanford_alpaca, 2023.
- Team, Q. Qwen2 technical report, 2024a.
- Team, Q. Qwen2.5: A party of foundation models, September 2024b. URL <https://qwenlm.github.io/blog/qwen2.5/>.
- Thorne, J., Vlachos, A., Cocarascu, O., Christodoulopoulos, C., and Mittal, A. The FEVER2.0 shared task. In *Proceedings of the Second Workshop on Fact Extraction and VERification (FEVER)*, 2018.
- Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023a.
- Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023b.

Turner, A. M., Thiergart, L., Leech, G., Udell, D., Vazquez, J. J., Mini, U., and MacDiarmid, M. Steering language models with activation engineering, 2024. URL <https://arxiv.org/abs/2308.10248>.

Wu, W., Morris, J. X., and Levine, L. Do language models plan ahead for future tokens? *arXiv preprint arXiv:2404.00859*, 2024.

Zhou, Z., Liu, Z., Liu, J., Dong, Z., Yang, C., and Qiao, Y. Weak-to-strong search: Align large language models via searching over small language models. *arXiv preprint arXiv:2405.19262*, 2024.

Zou, A., Phan, L., Chen, S., Campbell, J., Guo, P., Ren, R., Pan, A., Yin, X., Mazeika, M., Dombrowski, A.-K., et al. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023.